



School of Communication, Languages and Performing Arts
Department of Communication

August 2017 Semester
COM 302X: Statistics for Communication Research
Final Exam

Instructions

1. This paper has **FOUR** questions.
2. Answer Question **ONE** and any other **TWO** questions
3. The exam is two hours long.
4. Any form of cheating will lead to immediate dismissal from the exam room.
The candidate will earn a failing grade.
5. Please show all your working clearly.

EXAM BEGINS ON SECOND PAGE.

QUESTION ONE (30MKS)

- a) Distinguish between the following statistical concepts
- i) Measures of central tendency and Dispersion (2mks)
 - ii) Type I and Type II error (2mks)
 - iii) Continuous and Discrete variable (2mks)
 - iv) Sample and Population (2mks)
 - v) Class interval and class limit (2mks)
- b) The following data shows the marks the student scored in an IQ test interview done to students who had applied for a public relation job in one Media Company. Study the table below and answer the question that follows

Classes (marks)	20-30	30-40	40-50	50-60	60-70	70-80
Frequencies(No of students)	4	4	8	6	3	2

Calculate the following:

- i) Standard deviation (6mks)
 - ii) Median using the formulae..... (4mks)
 - iii) Quartile deviation (6mks)
- c) Discuss three types of correlation analysis (4mks)

QUESTION TWO (20MKS)

Students at Daystar University in the 302 evening class were polled by a researcher attempting to establish a relationship between hours of study in the week immediately preceding the end of semester exam and the marks received on the exam. The surveyor gathered the data listed in the accompanying table.

Hours of study	25	12	18	26	19	20	23	15	22	8
Exam score	93	57	55	90	82	95	95	80	85	61

- i) Find the least squares regression line by identifying the appropriate dependent and independent variable. (10mks)
- ii) Interpret the meaning of the values of β_0 and β_1 calculated in part (i). 4mks

- iii) Compute the correlation of coefficient and coefficient of determination and interpret them. (6mks)

QUESTION THREE (20MKS)

- (a) Explain the main characteristics of the normal distribution curve (5marks)
- (b) A supermarket stocks groceries whose prices are normally distributed with a mean of Ksh 200 and a standard deviation of 15. On a certain weekend Akinyi had visitors and went shopping for groceries. She spent Ksh 5000 on her shopping.
- i) How much did she spend on groceries whose prices are less than Ksh 220 ($X < 220$)? (2marks)
- ii) How much did she spend on groceries prices are more than Ksh 150 ($X \geq 150$)? (3marks)
- iii) How much did she spend on groceries whose prices are between Ksh 180 and Ksh 210? ($180 \leq x \leq 210$). (4marks)
- (c) Explain three advantages of sampling (6marks)

QUESTION FOUR (20MKS)

- a) Explain two advantages of median as a measure of central tendency (4mks)
- b) Explain four steps in hypothesis testing (6mks)
- c) The number of parts for a particular spare part in a factory was found to vary from day to day. In sample study the following information was obtained.

Day	No of parts demanded
Monday	1124
Tuesday	1125
Wednesday	1110
Thursday	1120
Friday	1126
Saturday	1115
Total	6720

Using chi-square, test the hypothesis that the number of parts demanded does not depend on the day of the week at 5% level of significance (10mks)